

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

1st April 2021

**Subject CP1 – Actuarial Practice
(Paper B)**

Time allowed: 3 Hours 30 Minutes (14.30 - 18.00 Hours)

Total Marks: 100

Your solution should link to the information provided in the questions. Limited credit will be given to solutions where generic answers have been given that do not refer back to the information provided.

Assignment 1

Background information

A coastal area in a developing country is prone to frequent flooding due to heavy rains. The last flooding was in year 2018 and since many residents had not purchased flood insurance they had to face significant uninsured losses after the flood.

Q. 1) Suggest reasons why many people in the region may not have purchased any insurance policy for coverage against flood perils. [9]

Q. 2) Discuss the actions that the Government could take to protect its citizens from flood risks

- a)** Through its direct intervention (4)
- b)** Through the insurance industry (4)

[8]

A large private composite insurer in the country is developing a flood insurance product which would also be offered to the people living in flood prone regions.

Q. 3) Discuss what steps the insurance company could take to encourage people to buy the flood insurance product. You should discuss the following aspects in your answer-

- a)** Product design (5)
- b)** Product sales and marketing (4)

[9]

Q. 4) List the rating factors that could be used in pricing a flood insurance product. [5]

Q. 5) As the Actuary to the Life Division of the company, outline reasons why you would like to perform analysis of surplus. [4]

The results of analysis of surplus show losses from surrenders from the Company's regular pay non-participating endowment product. This trend has been observed for the past few years.

Q. 6) Outline the factors that need to be considered when setting the surrender value for a regular pay non-participating endowment product? [5]

Q. 7) You have been asked to prepare a note analysing the reasons for losses arising from the surrender experience. Discuss the various analysis/investigations that you need to perform [10]
[50]

Assignment 2

Background information

A large general insurer, XYZ GI Company Limited, in a developed country, has a sizable market share and currently offers traditional general insurance products. It is listed in the stock market. Recently, there has been an entry of lot of new age players in the market, who have come up with unique products and solutions, competitive premiums and they are slowly eating into the market share of XYZ.

Over past few years there has been a surge in Cab/taxis offered by aggregators in the country where a customer books a cab via mobile applications. XYZ GI plans to collaborate with the biggest cab aggregator to offer a unique insurance product.

Proposed coverages under the product are as below:

- Accidental Cover: A fixed amount of INR 5,00,000 will be payable on death of each rider/s due to accident that occurs during travel period
- Permanent disability cover: If a rider/s become totally and permanently disabled due to accident that occurs during travel period then a fixed benefit of INR 5,00,000 each would become payable
- Hospitalization Cover: Reimbursement of expenses incurred by rider/s on hospitalization due to accident occurring during travel period subject to a maximum limit of INR 5,00,000 each
- Trip Delay: Bill will be waived if a rider's journey is delayed by more than 1 hour from the scheduled arrival time committed at the time of making a booking.
- Replacement costs in case of irreversible damage to 1 laptop belonging to riders due to an accident during a trip

All rides will be covered and premium will be paid by Cab Aggregator.

- Q. 8)** Why XYZ GI would be interested in offering such a product. [5]
- Q. 9)** What are the risks in the offering from the insurance company's perspective relating to-
- a) Benefit design (8)
 - b) Other risks (8)
- [16]
- Q. 10)** Discuss the various sources of internal and external data that the company could use in determining the risk premium for the various coverages. Outline the adjustments that would be required to be made to such sourced data. [12]
- Q. 11)** Discuss the factors that need to be considered when finalizing the office premium for the product. [6]

XYZ GI's Board is worried that the new startup insurance companies may poach people from its senior and middle management levels. The Board is considering the following scheme to retain its managers-

Each employee at the middle/senior management level will be awarded Employee stock options (ESO) at the time of annual performance review depending on his/her performance. ESOs are similar to regular call options and will give the employee the right to buy the company's shares at a specified price. The ESOs will have a vesting period of 5 years. After which, the option has to be exercised within 3 years otherwise it shall lapse. The ESOs are not tradeable in the market.

Q. 12) Discuss the pros and cons of the scheme for the employer.

[11]

[50]
